

AAST

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Embedded systems & IoT



PROJECT PROPOSAL

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1. Summary

The main purpose of the project is to build a fully automated Robot with a robotic arm that can play chess against a human player using a chess engine. The expected result of the project is to have a robotic arm with 6 degrees of freedom that can detect the user move with various ways on the chess board and perform the best move for the current situation. The total anticipated budget is 9000 EGP.

2. Introduction

The project is built in order to show how the technology and the world of robotics is starting to improve and that it can beat an experienced human and read the way he/she thinks in order to win. The project will use a Raspberry PI 4 B+ microcontroller. This microcontroller will be used to control a 3D printed robotic arm that will consist of 6 different motors. The 6 motors are divided into 2 groups which are the SG90 motor and the MG996r motor. Each of these motors will be used to control one of the arm joints in order to emulate the human hand movement with 6 degrees of freedom. As the arm will be able to move, now it is the time to play chess so a camera will be connected on the top of the arm so that the camera captures the chess board whenever the user presses on the switch that he/she played their move. The image captured is then passed to the microcontroller in order to be processed. Firstly, the user move captured by the camera will be visualized on the chess board shown on a web app that is designed for the project. The image of the board on the web app after the user move has been done is then passed to a chess engine that is connected to the web app. The chess engine will then process this image and take the best decision to perform for the current situation, the decision taken by the chess engine is then made on the web app. The microcontroller then takes the decision taken by the chess engine from the app and sends it as an instruction to the robotic arm to perform the same move on the board and then the user can play again. Before the game starts the user will be able to choose the difficulty of the game he/she wants to play according to their chess rating so that they can also compete with the robot not just lose. One of the project most beneficial ideas is that the project will allow the user to play using voice control, but why? A lot of people are talented and they know how to play the game professionally but due to any kind of disability they can not do so. That's why we decided to add the voice control feature so that the user can give the move they want through a microphone and the robotic arm will move the piece that the user wanted to the targeted place without the need that the user should move it on his own. Then the same steps explained previously are repeated so that the robot can do his own move against the player. The robot will also be able to interact with user so that when the user performs a good move the robot will give a voice output through a speaker such as "What a brilliant move!" or when the user performs a wrong move on the chess board the robot will give the user an alert that this move is incorrect so that the user can redo the move and then continue the game. Not only that but when the game ends the robot will shake his hands with the user after the game in terms of respect between players and then if the robot was the winner a small celebration will be done by the robotic arm. The project will be used in our university this term not only as a project but also as a competitor as our university campus is conducting a chess tournament and it is planned that when the project is built completely the winner of the competition will face our robot so that we can see who will win between AI and humans.

3. Needs/Problems

There are different kinds of problems to be faced throughout the project. Firstly, controlling the robotic arm and its degrees of freedom efficiently which is a problem that has never been addressed before by any of the group members. Secondly, the communication between the microcontroller, web app, and the chess engine in order to send and receive data to perform outputs. Lastly, in the voice control part there are a lot of restrictions and problems such as background noises that could affect the sound listened by the robot which would affect the efficiency of the robot.

4. Goals/Objectives

- Learn how to control a robotic arm and study all its restrictions to know how to make it move free so that it can play chess efficiently.
- Learn IoT and network communication between the microcontroller and the web server.
- Build an efficient microphone model that is capable to hear the user voice and filter the noises

5. Procedures/Scope of Work

First of all, a circuit schematic of the robotic arm is to be designed in order to test the arm movement and control by simulation. The pieces of the robotic arm will be 3D printed and then the circuit will be connected as the schematic. Before playing chess the arm will be tested to just hold objects and drop them in a small area so that we can fix any kind of errors in the arm before integrating the chess part. When the arm is ready, fully capable, and efficient of all the kinds of movements the chess engine and the camera will be connected to the robot so that we can start testing and training the robot to play chess and solve the new errors to be faced in the communication between all the project devices. Lastly, when the robot becomes a professional chess player “Magnus Armsen” the voice control part will be added so that the robot can also play instead of the user not just playing for his own.

6. Timetable

	Description of Work	Start and End Dates
Phase One	3D printing the robotic arm parts	19/03/2023 to 25/03/2023
Phase Two	Drawing the circuit schematic	26/03/2023 to 1/04/2023
Phase Three	Program the robotic arm to move	2/04/2023 to 8/04/2023
Phase Four	Integrate the robot with the chess engine	9/04/2023 to 22/04/2023
Phase Five	Voice control and voice outputs	23/04/2023 to 29/04/2023

Budget

	Component (Quantity)	Price
First	Raspberry PI 4 Model B+ (1)	5600 EGP
Second	Raspberry PI 4 adapter with ON/OFF switch (1)	120 EGP
Third	Raspberry PI heat sink kit (1)	30 EGP
Fourth	HDMI Female to Mini HDMI convertor (1)	15 EGP
Fifth	SanDisk 64GB Sd card (1)	150 EGP
Sixth	Speaker (1)	20 EGP
Seventh	Microphone (1)	20 EGP
Eighth	Arm Gripper (1)	150 EGP
Nineth	MG996r (3)	555 EGP
Tenth	SG90 (3)	240 EGP
Eleventh	Raspberry PI camera board – V2 8MP (1)	850 EGP
Twelfth	Raspberry PI camera ribbon flexible flat cable FFC 1.0mm 15Pin 100cm (1)	55 EGP
Thirteenth	Raspberry PI CSI camera holder (1)	65 EGP
Fourteenth	3D printer filament 1Kg (1)	530 EGP
Fifteenth	Extras (jumpers, 4mm bolts and nuts, screws, domain, chess board, etc.)	600 EGP
	Total	9000 EGP

7. Next Steps

The future work to be done on the project.

- Build a database that keeps records of all the matches played by the robot against a specific user and the match difficulty.
- The robot can detect if the user has performed a wrong move. For now, the robot will react to that by giving an alert but in future the robot will also be able to put that piece back to its previous position if the move was wrong.

8. Appendix

- <https://stockfishchess.org/Research materials>